Exercise 4
UNION & UNION ALL EXERCISE

01 Questions 01 – 05 (UNION)

Table : branch_sandton_accounts

accounts_id	account_holder
A001	Nomvula Dlamini
A002	David Mokoena
A003	Lerato Sithole
A004	Peter Nkosi

branch_rosebank_accounts.

account_id	account_holder
A003	Lerato Sithole
A004	Peter Nkosi
A005	Zanele Khumalo
A006	Thabo Motha

```
(SELECT `account_id`,
        `account_holder`,
 'Johannesburg' AS city
 FROM branch_sandton_accounts)
  UNION
(SELECT `account_id`,
        `account_holder`,
 'Johannesburg' AS City
 FROM branch_rosebank_accounts);
```

accounts_id	account_holder	city
A001	Nomvula Dlamini	Johannesburg
A002	David Mokoena	Johannesburg
A003	Lerato Sithole	Johannesburg
A004	Peter Nkosi	Johannesburg
A005	Zanele Khumalo	Johannesburg
A006	Thabo Motha	Johannesburg

02

Table : Savings_products

product_code	product_name
SV01	Basic Savings
SV02	Premium Savings
SV03	Youth Savings
SV04	Business Savings

Current_products

product_code	product_name
CR01	StandardCurrent
CR02	Gold current
SV03	Youth Savings
CR03	Business current

2.
```
(SELECT product_code,
        product_name,
        'Savings' AS product_type
 FROM savings_products)
 UNION
(SELECT product_code,
        product_name,
        'Savings' AS product_type
 FROM current_products);
```

Results table

product_code	product_name	product_type
SV01	Basic savings	savings
SV02	Premium savings	savings
SV03	Youth savings	savings
SV04	Business savings	savings.
CR01	Standard current	Savings
CR02	Gold current	savings
CR03	Business current	savings

3. Table: retail_banking_staff

staff_id	staff_name
S001	Mpho Radebe
S002	Brian Tshabalala
S003	Aisha Patel
S004	Kabelo Moabelo

Table: corporate_banking_staff

staff_id	staff_name
S003	Aisha Patel
S005	Nandi dube
S006	Sipho khumalo
S004	Kabelo Moabelo

③
```
(SELECT 'staff_id',
        'staff_name',
        'email' AS email
 FROM retail_banking_staff)
  UNION
(SELECT 'staff_id',
        'staff_name',
        'Email' AS email
 FROM corporate_banking_staff.
```

Results table

staff_id	staff_name	Email
S003	Aisha Patel	Email
S005	Nandi dude	Email
S006	Sipho Khumalo	Email
S004	Kabelo Moabelo	Email
S001	Mpho Radebe	Email
S002	Brian Shabalala	email.

4. Table: mobile_branch_cities

City_code	city_name
C01	Johannesburg
C02	Pretoria
C03	CapeTown
C04	Durban

Table: digital_branch_cities

city_code	city_name
C03	Cape Town
C05	Polokwane
C06	Port Elizabeth
C01	Johannesburg

④
```
(SELECT 'city_code',
        'city_name',
        'region' AS region
FROM mobile_branch_cities)
 UNION
(SELECT 'city_code',
        'city_name',
        'region' AS region
FROM digital_branch_cities);
```

Results_table.

City_code	City_name	Region
C01	Johannesburg	region
C02	Pretoria	region
C03	Cape Town	region
C04	Durban	region
C05	Polokwane	region
C06	Port Elizabeth	region

⑤ Table: push_notifications_targets

customer_id	customer_name
C001	Nomsa Zwane
C1002	Andile Buthelezi
C1003	Fatima Mahomed
C1004	Ryno van Zyl

in_app_banner_targets

Customer_id	customer_name
C1003	~~Fatima Mohamed~~ ~~Thandeka Cele~~
C1005	Thandeka Cele
C1006	Samuel Nkosi
C002	Andile Buthelezi

(5)
```
(SELECT `customer_id`,
        `customer_name`,
        'Notification' AS segment
 FROM push_notifications_target)
 UNION
(SELECT `customer_id`,
        `customer_name`,
        `Notification` AS segment
 FROM inapp_banner_targets);
```

Results table

customer_id	customer_name	Segment
C1001	Nomsa Zwane	Notification
C1002	Andile Buthelezi	Notification
C1003	Fatima Mahomed	Notification
C1004	Ryno van Zyl	Notification
C1005	Thandeka Cele	Notification
C1006	Samuel Nkosi	Notification

(6) Questions 6–10 (UNION ALL)

Table atm01_transactions

txn_id	account_id	Amount
T1001	A001	500.00
T1002	A002	1200.00
T1003	A001	300.00
T1004	A003	750.00

Table : atm02_transactions

txn_id	account_id	amount
T1003	A001	300.00
T1005	A004	900.00
T1006	A002	450.00
T1007	A005	150.00

Transaction_id , account_id , amount, transaction_date.

```
(SELECT 'txn_id',
        'account_id',
        'amount',
        'Date' AS transaction_date
        FROM atm01_transaction)
 UNION ALL
(SELECT 'txn_id',
        'account_id',
        'amount',
        'Date' AS transaction_date
  FROM atm02_transactions);
```

Results

txn_id	account_id	amount	Transaction_date
T1001	A001	500	Date
T1002	A002	1200	Date
T1003	A003	300	Date
T1004	A003	150	Date
T1003	A001	300	Date
T1005	A004	900	Date
T1006	A002	450	Date
T1007	A005	150	

The gauteng_loan_applications

app_id	customer_id	Loan_type
LA001	C1001	Home loan
LA002	C1002	Vehicle loan
LA003	C1003	Personal
LA004	C1004	Home loan

western_cape_loan_applications

app_id	customer_id	Loan_type
LA003	C1003	Personal
LA005	C1005	Vehicle loan
LA006	C1006	Business
LA007	C1007	Home loan.

```
(SELECT `app_id`,
        `customer_id`,
        `loan_type`,
        'Amount' AS amount_requested
 FROM gauteng_loan_applications)
   UNION ALL
(SELECT `app_id`,
        `customer_id`,
        `loan_type`,
        'Amount' AS amount_requested
 FROM westerncape_loan_applications);
```

Results

app_id	customer_id	loan_type	amount_requested
LA001	C1001	Home loan	Amount
LA002	C1002	Vehicle loan	Amount
LA003	C1003	Personal loan	Amount
LA004	C1004	Home loan	Amount
LA003	C1003	Personal	Amount
LA005	C1005	Vehicle	Amount
LA006	C1006	Business	Amount
LA007	C1007	Home loan	Amount.

8. Table : email_complaints

Complaint_id	customer_id
EC001	C2001
EC002	C2002
EC003	C2003
EC004	C2004

Table : app_complaints

Complaint_id	Customer_id
AC001	C2005
AC002	C2001
AC003	C2006
AC004	C2007

```
(SELECT 'complaint_id',
        'customer_id',
        '2026-10-07' AS logged_date,
        'Email' As category
 FROM email_complaints )
  UNION ALL
(SELECT 'complaint_id',
        'customer_id',
        '2026-10-15', AS logged_date,
        'App' AS category
 FROM app_complaints );
```

Results table

Complaint_id	Customer_id	logged_date	category
AC001	C2005	2026-10-15	App
AC002	C2001	2026-10-15	App
AC003	C2006	2026-10-15	App
AC004	C2007	2026-10-15	App
EC001	C2001	2026-10-07	Email
EC002	C2002	2026-10-07	Email
EC003	C2003	2026-10-07	Email
EC004	C2004	2026-10-07	Email

09 Table: april_payments

payment_id	account_id	amount
PAY001	A001	1250
PAY002	A002	4800
PAY003	A003	9200
PAY004	A004	3800

Table: may_payments

Payment_id	account_id	amount
~~PA005~~ PAY005	A001	12500.
PA006	A005	7600
PAY007	A002	5100
PAY008	A006	2800

```
(SELECT `payment_id`,
        `account_id`,
        `amount`,
        '2026-04-01' AS payment_date
   FROM april_payments)
UNION ALL
(SELECT `payment_id`,
        `account_id`,
        `amount`,
        '2026-05-01' AS payment_date
   FROM may_payments);
```

Results Table

payment_id	account_id	amount	payment_date
PAY001	A001	1250	2026-04-01
PAY002	A002	4500	2026-04-01
PAY003	A003	9200	2026-04-01
PAY004	A004	3300	2026-04-01
PAY005	A001	12500	2026-05-01
PAY006	A005	7600	2026-05-01
PAY007	A002	5100	2026-05-01
PAY008	A006	2800	2026-05-01

Table

10. debit_entries account_id

entry_id	account_id	entry_type
DR001	A001	Debit
DR002	A002	Debit
DR003	A003	Debit
DR004	A004	Debit

Table: credit_entries

entry_id	account_id	entry_
CR001	A001	Credit
CR002	A005	Credit
CR003	A002	Credit
CR004	A006	Credit

```
(SELECT 'entry-id',
        'account_id',
        'entry_type',
        'Amount' AS Amount,
        '2025-12-01' AS entry_date
  FROM debit_entries)
UNION ALL
(SELECT 'entry_id',
        'account-id',
        'entry_type',
        'Amount' AS amount
        '2025-09-01' AS entry_date
  FROM credit_entries);
```

Results table

entry_id	account_id	entry_type	Amount	Entry_date
DR001	A001	Debit	Amount	2025-12-01
DR002	A002	Debit	Amount	2025-12-01
DR003	A003	Debit	Amount	2025-12-01
DR004	A004	Debit	Amount	2025-12-01
CR001	A001	Credit	Amount	2025-09-01
CR002	A005	Credit	Amount	2025-09-01
CR003	A002	Credit	Amount	2025-09-01
CR004	A006	Credit	Amount	2025-09-01

Bonus Questions section 3

(a) UNION – Because we want a unique list, and union removes duplicates.

(b) UNION ALL – Because they want to count everyones transaction that passed.

(c) Because the columns are not storing the similar kind of data, therefore not compatible.